

Chapter 11, Problem 2PP

Problem

A current $\mathbf{I} = 33\angle 30^\circ$ A flows through an impedance $\mathbf{Z} = 40\angle -22^\circ \Omega$. Find the average power delivered to the impedance.

Step-by-step solution

Step 1 of 2

Consider the following data:

Current, $\mathbf{I} = 33\angle 30^\circ$ A.

Impedance, $\mathbf{Z} = 40\angle -22^\circ \Omega$.

Calculate the voltage across the impedance:

$$\begin{aligned}\mathbf{V} &= \mathbf{IZ} \\ &= (33\angle 30^\circ \text{ A})(40\angle -22^\circ \Omega) \\ &= 1320\angle 8^\circ \text{ V}\end{aligned}$$

Step 2 of 2

Calculate the average power P delivered to the impedance:

$$\begin{aligned}P &= \frac{1}{2} V_m I_m \cos(\theta_v - \theta_i) \\ &= \frac{1}{2} (1320)(33) \cos(8^\circ - 30^\circ) \\ &= 20.194 \text{ kW}\end{aligned}$$

Therefore, the value of average power P delivered to the impedance is 20.194 kW.